1.7.4 Square root of decimal numbers

To compute the square root of numbers in the decimal form, we simply follow the following steps:

Step 1:

Make even number of decimal places even by affixing a zero on the extreme right of the digit in the decimal part (only if required).

To find $\sqrt{42.25}$

Step 2:

The number has an integral part and a decimal part. In the integral part, mark the bars as done in the case of division method to find the square root of a perfect square number.

We put the bars as $\sqrt{\overline{42}.\overline{25}}$

Step 3:

In the decimal part, mark the bars on every pair of digits beginning with the first decimal place.

Step 4:

Now, calculate the square root by long division method.

$$\begin{array}{r|l} & \;\;6\;\;\;5 \\ \hline 6 & \overline{42}\;\overline{25} \\ & 36 \\ \hline 125 & \;\;6\;25 \\ & \;\;6\;25 \\ \hline & \;\;\;\;0 \end{array}$$

Step 5:

Put the decimal point in the square root as soon as the integral part is exhausted. $\therefore \sqrt{42.25} = 6.5$

1.7.5 Square root of product and quotient of numbers

For any two positive numbers *a* and *b*. we have

(i) $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ and (ii) $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b \neq 0)$

Example 1.28

Find the value of $\sqrt{256}$.

Try these

Find the square root of

1. 5.4756 2. 19.36 3. 116.64

Solution:

$$\sqrt{256} = \sqrt{16 \times 16} = \sqrt{16} \times \sqrt{16} = 4 \times 4 = 16 \text{ (or) } \sqrt{256} = \sqrt{64 \times 4} = \sqrt{64} \times \sqrt{4} = 8 \times 2 = 16.$$

Think

Try to fill in the blanks using $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$.

$\sqrt{36} = 6$	$\sqrt{9} \times \sqrt{4} = 3 \times 2 = 6$	Is $\sqrt{36} = \sqrt{9} \times \sqrt{4}$?	$\sqrt{81} = ?$	$\sqrt{9} \times \sqrt{9} =$ __ × __ = __	Is $\sqrt{81} = \sqrt{9} \times \sqrt{9}$?
$\sqrt{144} = ?$	$\sqrt{9} \times \sqrt{16} =$ __ × __ = __	Is $\sqrt{144} = \sqrt{9} \times \sqrt{16}$?	$\sqrt{144} = ?$	$\sqrt{36} \times \sqrt{4} =$ __ × __ = __	Is $\sqrt{144} = \sqrt{36} \times \sqrt{4}$?
$\sqrt{100} = ?$	$\sqrt{25} \times \sqrt{4} =$ __ × __ = __	Is $\sqrt{100} = \sqrt{25} \times \sqrt{4}$?	$\sqrt{1225} = ?$	$\sqrt{25} \times \sqrt{49} =$ __ × __ = __	Is $\sqrt{1225} = \sqrt{25} \times \sqrt{49}$?

Activity

Attempt to prepare a table of square root problems as in the above case to show that if *a* and *b* are two perfect square numbers, then $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b \neq 0)$. We can use this idea to compute certain square-root problems easily.